

Stack 07 audit candidate — ERC-7730 and streamed EIP-712

1. Identity and finite scope

Owner: Codex /root, vault agent-2. Main-worktree claim: docs/release/audit-claims/715-07/OWNER.md. Canonical master/SOP hashes are preserved in the validation bundle's canonical-doc-hashes.txt. Initial master hash was 67cae5f586d17212874072019de8d434e16a79b4edbc52c6e1b7d338d0b87a1. The main documents were reread before this freeze.

Immediate candidate predecessor: Stack 06 PR #859, audit/715-stack06-agent1-reconcile at 8ed653517a63278891353bcf939a9c3108feacb2. Frozen implementation head: 34a36e700ec31b7338b86eeb82e0c23a75da30bb. Inventory generation predecessor: e46aae5fd9804595dc492752ce6c2b06f9da6e86. Final containing head and hosted CI result belong in PR #837's description, because this report cannot embed its own commit hash. The owner explicitly authorized retargeting existing PR #837 and force-with-lease after validation and the full retrospective. No further Copilot request is authorized.

The historical Stack 07 commit 4695a11444fa6a3bb990e59d000b8149bc5b74f3 was replayed as 632de4f7f onto the predecessor intake, then reconciled through merges 66dc2df72 and 34a36e700. Its historical base b771669ba842d83790a69b6132f63cea82e5a861 is not the corrective candidate's base. Historical source inventory: 38 files, 10046 additions, 207 deletions. Per-commit file/count provenance is in 715-07-history-20260924.tsv; overlapping history counts must not be summed as the net current diff.

This block adds a bounded authenticated runtime-catalog/ABI/display subset and device-driven streamed EIP-712. Ordinary fully disclosed streamed EIP-712 works without AdvancedMode. Runtime-certified annotations and blind precomputed typed hashes require AdvancedMode. Runtime annotations remain additive to ordinary review and carry explicit source/warning/intent screens. Full ERC-7730 v2 conformance, firmware-trusted registry authority and later stack features are excluded. BTC-only omits these engines.

2. Source reconciliation and behavior coverage

Paths	Prior behavior; candidate behavior; why and evidence
eip712_stream.h/.c and native tests	Adds bounded schema discovery, canonical type/value hashing, visible leaf review and resumable typed signing. Repairs header/count bounds, nested dimensions and schema consistency. Fixed schemas and signed domain constraints survive repeated field walks. Published type-hash vectors, malformed encodings, array/schema negative controls and host repeated-field tests execute these properties.
erc7730_catalog.h/.c and tests	Adds streaming envelope/program validation and runtime-key authentication. Signed identity uses the entire program leaf. Canonical empty root and domain records are supported. Real signature tampering tests refute header/key bypass; source alias is the user-approved runtime alias.
erc7730_abi.h/.c, erc7730_abi_stream.h/.c and tests	Adds bounded recursive canonical ABI parsing. Accepts selector-only root while rejecting malformed graphs; stream owns its program view, eliminating caller-lifetime dependency. Chunk-boundary, offset/padding, caller-view overwrite and extreme-index tests run.
erc7730_program.h/.c, condition, format and tests	Adds bounded section/literal/path/condition/formatter readers. Domain operations are retained across chunks; unsupported shapes and oversized/missing values fail closed. Native typed selection/format and authenticated host fixtures cover execution.
erc7730_workflow.h/.c, erc7730_tx.h/.c and tests	Orchestrates definition replay, capture and signing continuation. SHA-256 binds calldata and typed domain/message digests across all passes; every typed display field advances. Negative controls catch removed commitments; host altered-signing-pass and multi-field fixtures pass.
ethereum.h/.c, fsm_msg_ethereum.h and Ethereum tests	Integrates runtime annotation plus mandatory ordinary review and restores direct chain-ID maximum validation. Adds runtime source/fingerprint/warning/intent on both calldata and typed paths. Real signatures match ordinary signing; decline/mutation paths return no signature.

Paths	Prior behavior; candidate behavior; why and evidence
fsm.c and fsm tests	Larger response union and phase-specific admission for new messages. Accepted continuations preserve active workflow; abort/session boundaries clear EIP-712 and ERC state; valid progress renews auto-lock while polling does not. Actual receive-path timer and session-boundary tests run.
board messages.h/.c and messagemap.def	Dense message lookup and new full-product protocol registration. Shared receive/tiny-buffer regressions run in both variants; BTC excludes Ethereum handlers.
firmware/unit CMakeLists.txt	Registers all new engines/tests only in the full product. Both native and ARM links pass; both product boundaries are executed.
deps/python-keepkey	Report/test-only correction 1d42a1f: screenless unlimited-approval refusal, accurate additive-review prose, and immediate wire Failure expectation. Actual hosted frames pass the corrected audit; seven focused tests pass.
ci.yml and scripts/emulator/python-keepkey-tests.sh	Removes only the now-implemented erc7730-runtime-review staged exclusion. Preserves predecessor's workflow-wide inventory and adds seven authenticated Stack 07 contracts to CI. Workflow lint, shell syntax and report validators pass.
scripts/emulator/test_stack07_regressions.py	New real signed-envelope, cancellation, source/intent, every-field, domain-substitution and signing-replay regression suite. Full seven pass; BTC seven intentionally skip because the product excludes Ethereum.
HANDOFF-ERC7730-ALPHA.md and audit documents	Historical alpha design is clearly labeled; bounded 7.15 contract and actual policy are stated. Intake retains historical failures. Current report, full retro, skip ledger and evidence supersede pending implementation statements in that intake.

3. Findings and security closure

The accompanying 715-07-first-copilot-retrospective-20260924.md/PDF is the complete finding ledger, cross-phase contract table, separate falsification pass and root-cause analysis. It dispositions all 13 inline comments plus the review body: nine fixed comments/eight roots, two overlapping policy corrections, two technical refutations. It also records five additional local/integration defects repaired. No known in-scope code defect remains after these local checks. GitHub thread closure is recorded in the PR after pushed fixes and individual replies.

The security matrix names host bytes/schema/domain, runtime authority, visible consent, bounds, timers and product separation; it follows them through parsing, replay, OLED and signing. Four deliberately removed protections produce four failing native tests; source bytes are restored and final native tests pass. A green aggregate count is not used as a substitute for these assertions.

4. Executed validation and limits

Check	Result and evidence
Full native	598 firmware, 19 board, 18 crypto and 72 Zcash crypto pass; final-native-full*.xml.
BTC native	134 firmware, 19 board and 18 crypto pass; final-native-btc*.xml.
Full host	739 pass, 92 skip; final-host-full.xml. Matching final emulator and pinned host.
BTC host	347 pass, 484 skip; final-host-btc.xml. Matching final emulator.
Authenticated Stack 07 contracts	Seven full-product tests pass; final-contracts-full.xml. BTC explicitly skips seven; final-contracts-btc.xml. Earlier passing OLED run in stack07-contracts-fixed.xml and stack07-oled-fixed.
Negative controls	Four expected failures in negative-controls.xml/log, normal restoration verified byte-for-byte. Native suite then passes.
ARM/resource	Both MinSizeRel products link. Full reserve 18900 bytes, BTC 39140; largest frame 7664 in both, exceeding the required 4096-byte margin. final-sram-full/btc.log and stack-usage archives.

Check	Result and evidence
OLED	Runtime alias/fingerprint, NOT verified by KeepKey warning, intent and both typed values visually inspected. Cancellation fixtures cover source, warning, intent and each field; intent-only fixture adds source/intent to ordinary review.
Report/automation	Both combined JUnit SECTIONS validations pass, six validator tests pass, actionlint and shell syntax pass, git diff --check passes. Code-bearing preflight is repeated against the final frozen range.

All 576 broad host skips are named with their actual reasons in 715-07-skips-20260924.tsv. The explicit staged capability list is evm-max-amount-review, evm-unknown-token-review, entropy-audit-budget, osmosis-wire-guards, hive-release-review, storage-v19-kdf, solana-lut-attestation. Those later-block capabilities are not claimed as covered. The ordinary broad suite and focused suites overlap and must not be summed as unique coverage. The first final host invocation used the wrong capability-variable name; its 13 full and one BTC failures are retained as incorrect-env-host-*.xml, then the exact workflow variable was used for the passing results. No assertion was weakened.

Earlier reused-flash/native failures, initial dispatch failures and infrastructure failures are explained in the full retrospective. Physical USB/controller timing, physical OLED/buttons, signed-device upgrade, complete release assembly and promotion remain unverified. The static SRAM gate is not a measured physical high-water mark. Hosted exact-head CI status is recorded separately in the PR, not inferred from local checks. Stack 06 acceptance remains under its owner's claim.

Evidence bundle: 715-07-validation-20260924.tgz; SHA256 manifest: 715-07-validation-sha256-20260924.txt. ARM images are retained externally with hashes, not included in Git. Builder: kktech/
firmware@sha256:7438e53933d47d53157ed6d96d864cb208597e62dce26235ace09d1063427fa2.

The first corrective hosted run 35962710863 failed full screenshot audit because the unlimited-approval refusal was incorrectly assigned two review frames. All 162 selected wire tests passed (26 skips), and BTC integration passed. Host pin 1d42a1f corrects the report and adds an exact immediate-Failure/no-ButtonRequest assertion. Seven focused tests and the corrected audit against the original hosted frames pass. These report/test-only changes do not invalidate earlier firmware/native/ARM or unchanged broad-host results; final hosted CI exercises the new host pin. See the added hosted-qualification section in the full retro.

5. Pins and adjacent inventory

One report/test-only host pin correction relative to 8ed653517a63278891353bcf939a9c3108feacb2: b745562be6a13cc83b2a547d7ef86637f4556dbb to 1d42a1fa401129da292a8b0fe4f6dc9bee727117, published on the BitHighlander host fork. It corrects the screenless unlimited-approval report contract and strengthens immediate-refusal assertions; no device code changed. Crypto: 8a392f70a5d5575ece3dfb35f115d4a4b27f497c; protocol: dd9c85dc747cf965fb0e7bf49615dc9e7568ee65; Python host: 1d42a1fa401129da292a8b0fe4f6dc9bee727117; GoogleTest: 7888184f28509dba839e3683409443e0b5bb8948; nested Ethereum list: 89a64f717e1690bb31adb3e4c38e23640357333c. The host fork/pin correction is inherited from Stack 06.

Immutable inventory command: git diff --numstat

8ed653517a63278891353bcf939a9c3108feacb2..e46aae5fd9804595dc492752ce6c2b06f9da6e86. Counts include all code, tests, workflows, documents and binary evidence. Binary counts are shown as dashes. The final containing head is in the PR body; inventory/path/count equality is checked after the report commit.

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Added Deleted Path
  1      1 .github/workflows/ci.yml
 89      0 docs/release/audit-units/715-07-first-copilot-retrospective-20260924.md
-      - docs/release/audit-units/715-07-first-copilot-retrospective-20260924.pdf
103      0 docs/release/audit-units/715-07-history-20260924.tsv
 62      0 docs/release/audit-units/715-07-intake-20260924.md
118      0 docs/release/audit-units/715-07-review-20260924.md
-      - docs/release/audit-units/715-07-review-20260924.pdf
577      0 docs/release/audit-units/715-07-skips-20260924.tsv
-      - docs/release/audit-units/715-07-validation-20260924.tgz
116      0 docs/release/audit-units/715-07-validation-sha256-20260924.txt
295      0 docs/security/HANDOFF-ERC7730-ALPHA.md
 43     24 include/keepkey/board/messages.h
210      0 include/keepkey/firmware/eip712_stream.h
 80      0 include/keepkey/firmware/erc7730_abi.h
 81      0 include/keepkey/firmware/erc7730_abi_stream.h
190      0 include/keepkey/firmware/erc7730_catalog.h
 20      0 include/keepkey/firmware/erc7730_condition.h
 19      0 include/keepkey/firmware/erc7730_format.h
296      0 include/keepkey/firmware/erc7730_program.h
 26      0 include/keepkey/firmware/erc7730_tx.h
153      0 include/keepkey/firmware/erc7730_workflow.h

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4      8 include/keepkey/firmware/ethereum.h
5     14 lib/board/messages.c
9      0 lib/firmware/CMakeLists.txt
1238   0 lib/firmware/eip712_stream.c
472    0 lib/firmware/erc7730_abi.c
433    0 lib/firmware/erc7730_abi_stream.c
1389   0 lib/firmware/erc7730_catalog.c
101    0 lib/firmware/erc7730_condition.c
196    0 lib/firmware/erc7730_format.c
836    0 lib/firmware/erc7730_program.c
46     0 lib/firmware/erc7730_tx.c
721    0 lib/firmware/erc7730_workflow.c
158    30 lib/firmware/ethereum.c
69     29 lib/firmware/fsm.c
965   106 lib/firmware/fsm_msg_ethereum.h
9      0 lib/firmware/messagemap.def
7      0 scripts/emulator/python-keepkey-tests.sh
189    0 scripts/emulator/test_stack07_regressions.py
9      0 unittests/firmware/CMakeLists.txt
764    0 unittests/firmware/eip712_stream.cpp
174    0 unittests/firmware/erc7730_abi.cpp
240    0 unittests/firmware/erc7730_abi_stream.cpp
678    0 unittests/firmware/erc7730_catalog.cpp
76     0 unittests/firmware/erc7730_condition.cpp
133    0 unittests/firmware/erc7730_format.cpp
390    0 unittests/firmware/erc7730_program.cpp
74     0 unittests/firmware/erc7730_tx.cpp
203    0 unittests/firmware/erc7730_workflow.cpp
13     0 unittests/firmware/ethereum.cpp
62     2 unittests/firmware/fsm.cpp

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6. Render/preflight and review checkpoint

Report source and PDF are frozen together; rendered pages are inspected for readable tables and matching inventory. Final path/count comparison, exact live base/head, code-bearing gate and whitespace check precede any future request. Implemented, targeted, integration and adversarial states are locally verified as listed. One historical external review delivered Changes recommended; this is not a current-head clean review. No second round is requested. The complete retro and pushed thread dispositions must be read before the owner considers another request. This receipt does not mark release acceptance or authorize merging/promoting the product.